

Ultracold interactions between Ba^+ and Li

Special issue: Magnetic field dependence

Albert-Ludwigs-Universität Freiburg

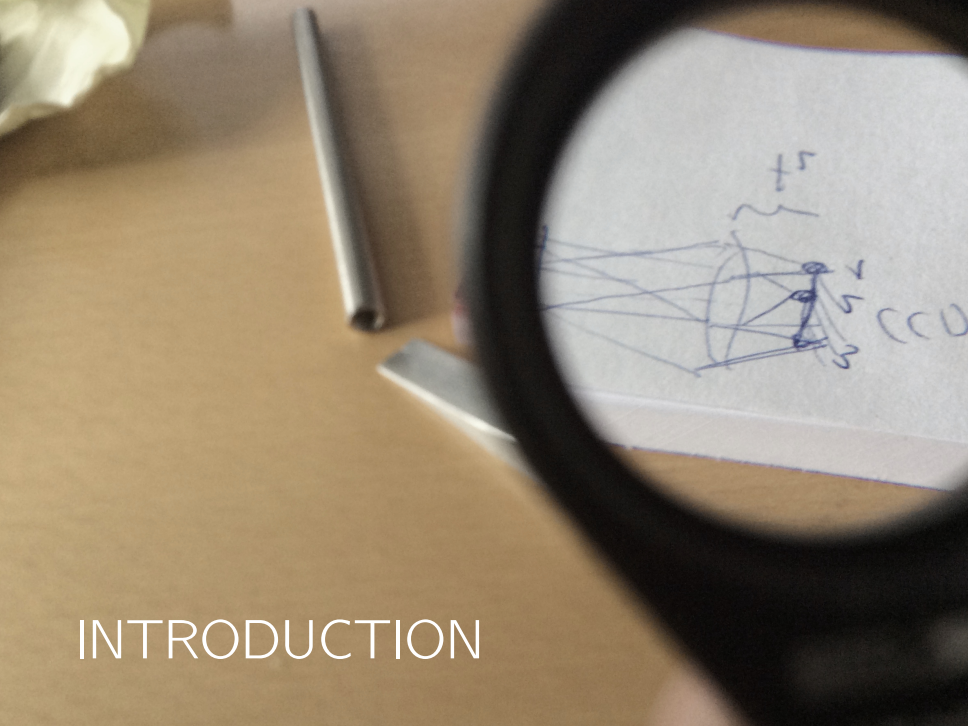


Fabian Thielemann, Pascal Weckesser, Daniel Hönig, Leon Karpa and
Tobias Schätz

Physikalisches Institut

16. Juni 2020

- Motivation & Recap
- Coherent control of spin-polarized Li
- B -Field dependent reactions
- Future experiments



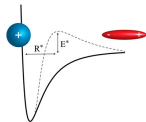
INTRODUCTION



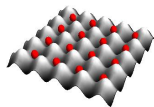
$$\propto \frac{1}{r}$$



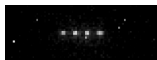
$$\propto \frac{1}{r^3}$$



$$\propto \frac{1}{r^4}$$



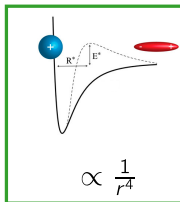
$$\propto \frac{1}{r^6}$$



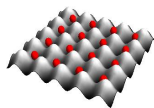
$$\propto \frac{1}{r}$$



$$\propto \frac{1}{r^3}$$

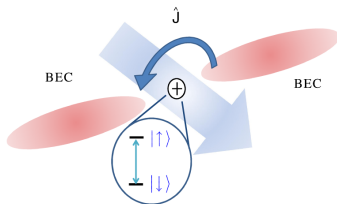


$$\propto \frac{1}{r^4}$$



$$\propto \frac{1}{r^6}$$

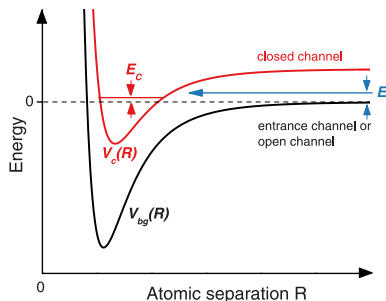
⇒ Long range, isotropic, many-body interactions



Gerritsma, R., et al. *Bosonic Josephson junction controlled by a single trapped ion*. Phys. rev. lett. 109.8 (2012).

Resonance scattering

A quasibound state coupling to the scattering state leads to a phase shift of $\approx \pi$ over a narrow range of energy.



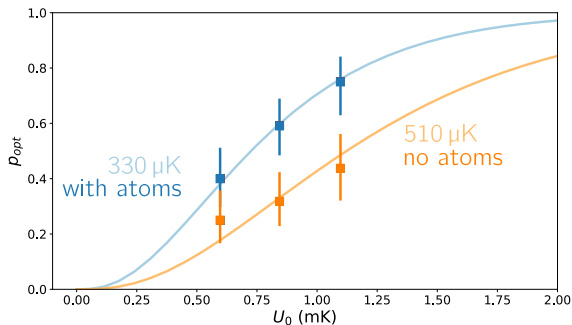
taken from: Chin et al. (2010), *Feshbach resonances in ultracold gases*

- Quasibound state in different potential manifold: *Feshbach resonance*
- Magnetic fields can be used to shift potentials relative to each other

$$a = a_{bg} \left(1 - \frac{\Delta}{B - B_0} \right)$$

Recap: State of experiment in '19

Sympathetic cooling of a single Ba^+ -ion in a bath of ^6Li

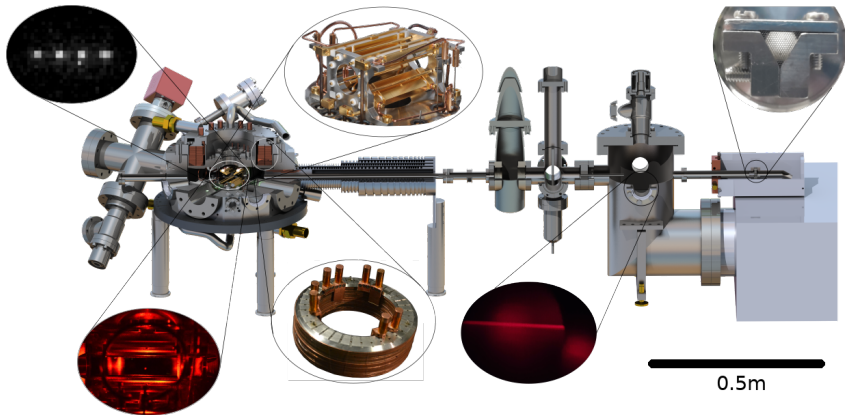


SETUP



Overview of the Experiment

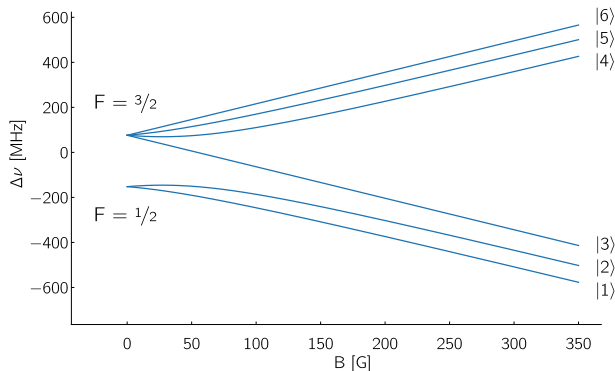
Single to tens of Doppler cooled Ba^+ -ions
and $\approx 2 \cdot 10^4$ ^6Li -atoms at $T < 700\text{ nK}$



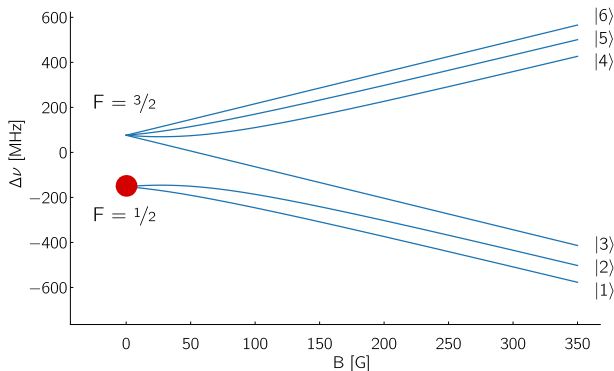
COHERENT CONTROL

Preparation of a spin-polarized atom cloud

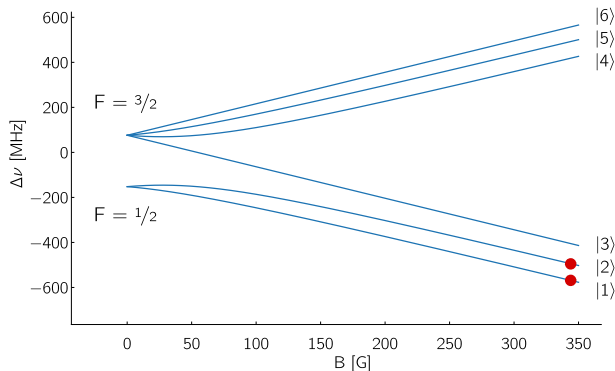
${}^6\text{Li}$: nuclear spin $I = 1$, D_2 -linewidth $\Gamma \approx 6\text{ MHz}$



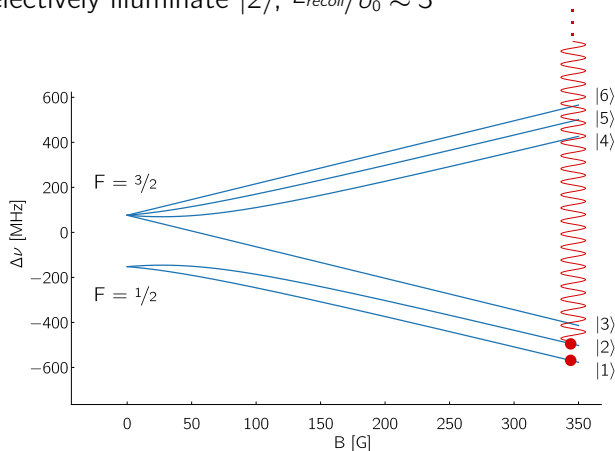
1) Optically pump atoms to $S_{1/2}(F = 1/2)$



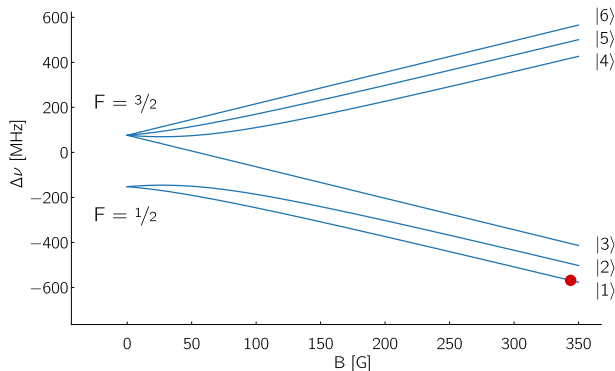
2) Evaporatively cool at high magnetic field to $T \approx 1\mu\text{K}$



3) Selectively illuminate $|2\rangle$, $E_{\text{recoil}}/U_0 \approx 3$

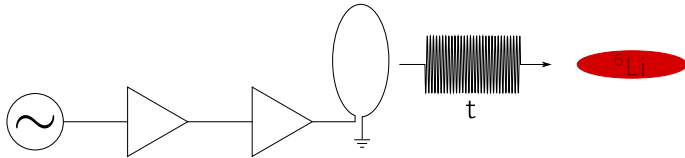


4) Around $N \approx 20\text{k}$ atoms in $|1\rangle$ left



Rabi oscillations - Li $|1\rangle \leftrightarrow |2\rangle$

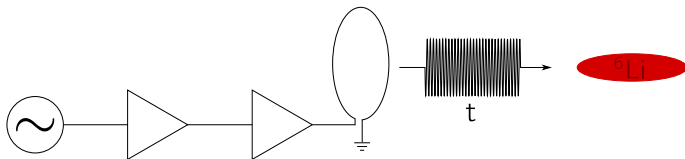
Interaction of two-level system with an oscillating electric field



Rabi oscillations - Li $|1\rangle \leftrightarrow |2\rangle$

Interaction of two-level system with an oscillating electric field

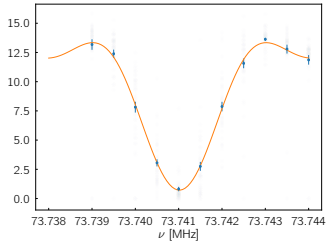
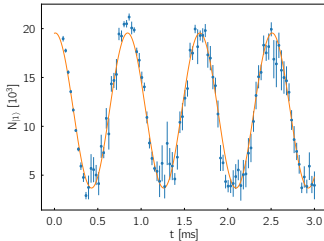
$$P_{|1\rangle} = \frac{\Omega^2}{\Omega'^2} \cos^2\left(\frac{\Omega' t}{2}\right) \quad \text{with} \quad \Omega' = \sqrt{\Omega^2 + \delta^2}$$



Rabi oscillations - Li $|1\rangle \leftrightarrow |2\rangle$

Interaction of two-level system with an oscillating electric field

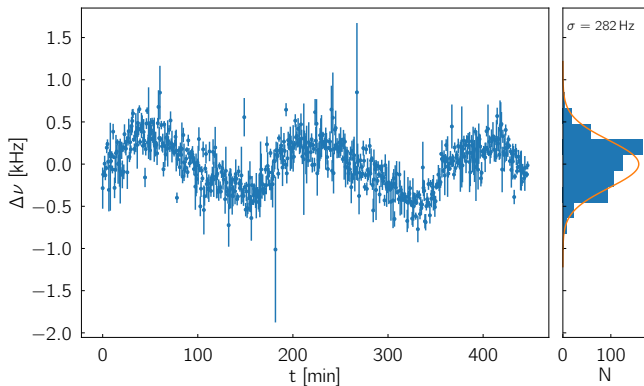
$$P_{|1\rangle} = \frac{\Omega^2}{\Omega'^2} \cos^2\left(\frac{\Omega' t}{2}\right) \quad \text{with} \quad \Omega' = \sqrt{\Omega^2 + \delta^2}$$



$$T_{\pi} = 440(10) \mu\text{s} \quad \nu_{12} = 73.74102(5) \text{ MHz}$$

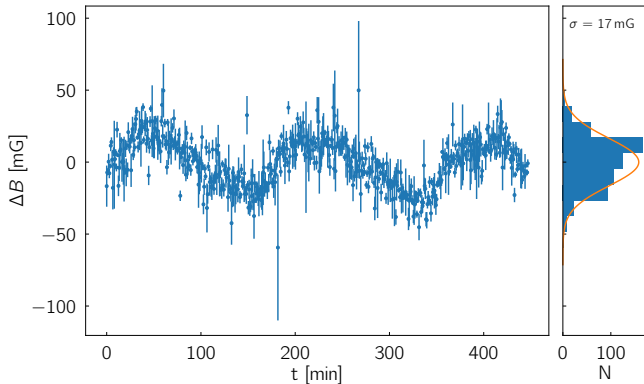
Relative long term stability of the hyperfine splitting at high magnetic fields

$$\frac{\sigma_{\nu_{12}}}{\nu_{12}} = 4 \cdot 10^{-6}$$



Attributing the frequency drift solely to the magnetic field, we get

$$\sigma_B = 17 \text{ mG at } \approx 300 \text{ G}$$

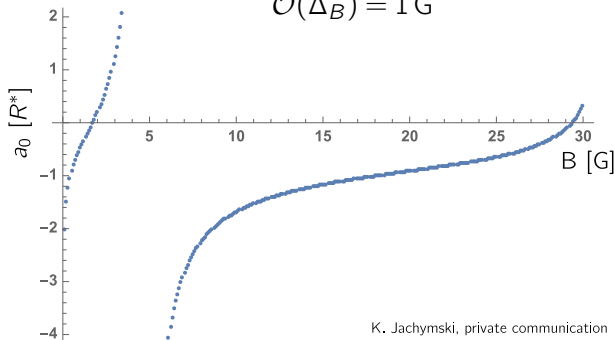


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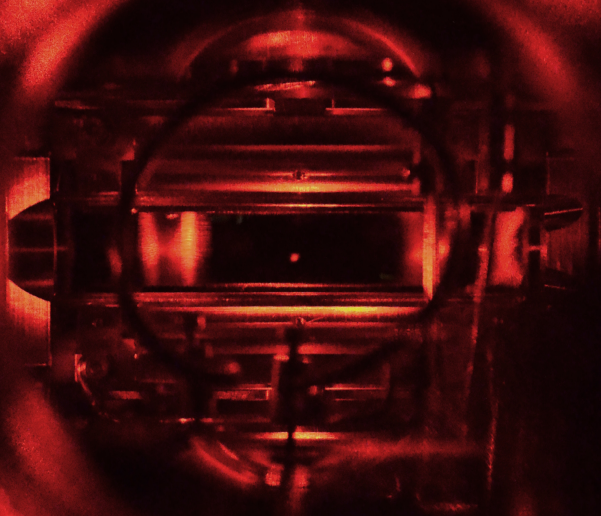
Predicted Feshbach resonances have

$$\mathcal{O}(\Delta_B) = 1 \text{ G}$$

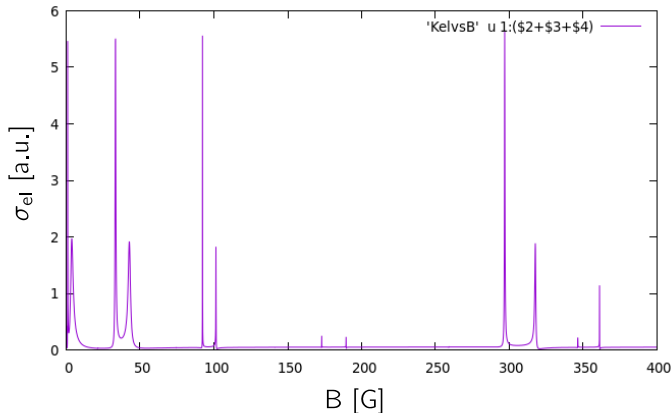


K. Jachymski, private communication

INTERACTION EXPERIMENTS



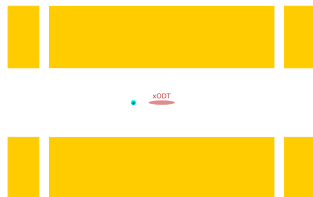
Poles in the scattering length a lead to pronounced peaks in the scattering cross section σ



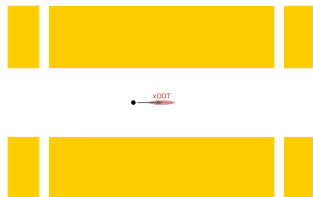
M. Tomza, private communication



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- 3 Overlap ion and atoms at magnetic field B for t_{int} .



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- 4 Detect Ba^+ -fluorescence on cooling transition.

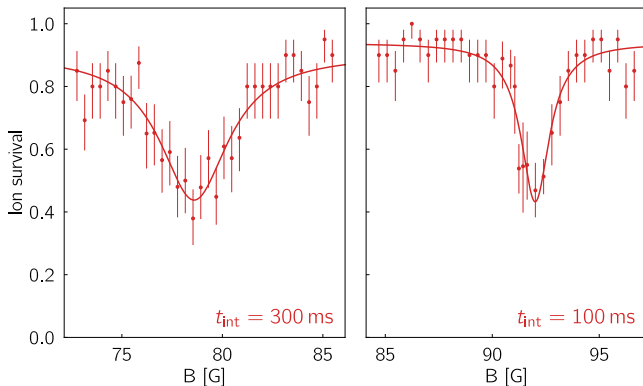


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- 2 Prepare $\approx 20\text{ k}$ ultracold ^6Li atoms in hyperfine state $|1\rangle$ or $|2\rangle$.
- 3 Overlap ion and atoms at magnetic field B for t_{int} .
- 4 Detect Ba^+ -fluorescence on cooling transition.
- 5 Repeat ad infinitum.

Two resonances:

$$B_0 = 78.6(1) \text{ G and } 92.0(1) \text{ G}$$

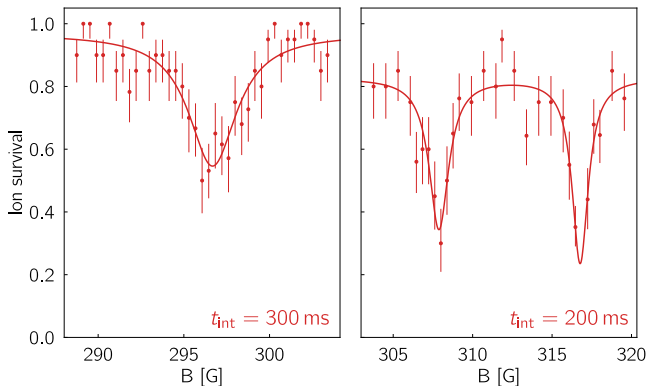
$$\text{FWHM} = 4.2(7) \text{ G and } 1.6(2) \text{ G}$$



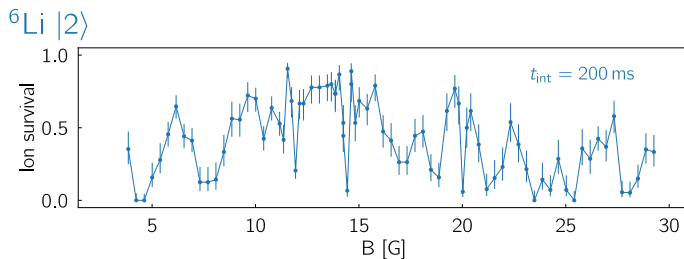
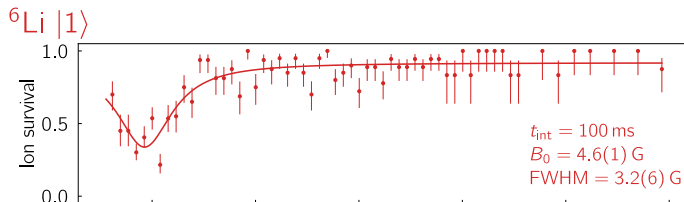
Three resonances:

$$B_0 = 296.7(1) \text{ G}, 307.9(1) \text{ G} \text{ and } 316.8(1) \text{ G}$$

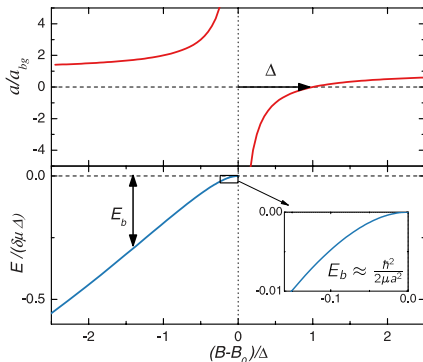
$$\text{FWHM} = 3.4(6) \text{ G}, 1.5(4) \text{ G} \text{ and } 1.2(4) \text{ G}$$



Comparison of entrance channels at low B



■ rf-spectroscopy near resonances



taken from: Chin et al. (2010), *Feshbach resonances in ultracold gases*

- rf-spectroscopy near resonances
- Characterize inelastic processes

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- Spin polarize Ba^+

- rf-spectroscopy near resonances
- Characterize inelastic processes
- Spin polarize Ba^+
- Transfer to a purely optical setup

Many thanks!



Pascal Weckesser, Daniel Hönig, Lennard Guth, Melissa Jaud, Leon Karpa, Tobias Schätz and the whole group!

Our collaborators: Michał Tomza, Olivier Dulieu, Krzysztof Jachymski



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